Adhesion & dwell-time gate NLRP3 priming & assembly in monocytes: a coupled multiscale CFD-ABM study

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Introduction

The NLRP3 inflammasome drives sterile vascular inflammation, and IL-1β inhibition lowers recurrent cardiovascular events [1]. Activation follows a two-signal route: NF-κB-dependent priming, then a danger signal [2]. How that architecture is engaged by the hemodynamics delivering monocytes to the endothelium is unknown. Transport and inflammasome models have advanced separately, and to our knowledge, no prior framework has carried hemodynamic particle trajectories into an agent-based model (ABM) encoding NF-κB-gated NLRP3 assembly. We tested whether priming is governed by local wall shear stress, the rolling-to-adhesion transition, or cumulative dwell time, using Lagrangian trajectories to drive an ABM of NF-κB and NLRP3 signaling.

Methods

Pulsatile flow in a patient-derived infrarenal aorta with aneurysmal dilation, from the Vascular Model Repository [3], was solved with OpenFOAM 10 pimpleFoam over two cardiac cycles (939,736 mesh cells, $y^+ < 1$, Carreau blood, Windkessel outlets at P_{mean} = 99 mmHg); cycle-to-cycle mean WSS varied by <1.5%. 9,991 monocyte parcels (d = 15 μm) were tracked one-way in the converged flow field with particleFoam, then imported into a BioDynaMo model [4] run for 168 h of biological time. Three monocyte subtypes were represented. NLRP3 accumulation from OxLDL, hypoxia and adhesion was gated by NF-κB exceeding a priming threshold of 0.3, encoding Signal 1, followed by cytokine secretion. NF-κB multipliers and Signal 2 coefficients were model parameters within biologically plausible ranges; OxLDL and oxygen were prescribed analytically. Ten seeded runs were performed, plus a variant freezing adhered cells in place. Varying the threshold, the adhered multiplier and the Signal 2 adhesion rate individually by ±20% changed peak adhesion count and time, and final NLRP3, by <12%.

Results

Peak adhesion occurred at 13.1 ± 1.5 h (88.6 ± 9.2 cells, CV 10.3%), with 57 ± 4% in the aneurysmal mid-segment (Figure 1A), 23 ± 3% in the inlet region and 20 ± 3% in the bifurcation region, consistent with flow deceleration and recirculation rather than WSS magnitude. In a representative run, rolling monocytes remained NLRP3-incompetent (0.000), NF-κB below threshold (0.064). Adhesion raised NF-κB 8.8-fold to 0.564, resulting in NLRP3=0.393. Transmigrating cells held the highest NF-κB, 0.944, but intermediate NLRP3, 0.563, since transmigration completes in 0.1 to 0.3 h, 10- to 30- fold shorter than the ~3 h assembly window. After hours of dual-signal exposure, in-tissue monocytes approached saturation (NLRP3=0.863, IL-1β=0.675; Figure 1B). The bifurcation region carried 1.63-fold NF-κB and 1.81-fold NLRP3 over the inlet region, consistent with axial OxLDL and hypoxia accumulation. Freezing adhered cells (two seeds) preserved the hierarchy and spatial ratios within ~10% but shifted peak count and timing by ~55% and ~30%.

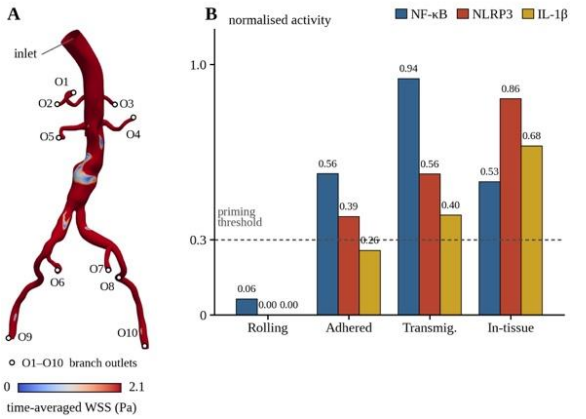


Figure 1: (A) Time-averaged WSS, patient-derived aorta. (B) Signaling by monocyte state at peak adhesion (representative run, t = 14.7 h); dashed line, priming threshold. NLRP3 tracks dwell time, not NF-κB.

Conclusions

Three testable hypotheses follow; only the second is emergent rather than prescribed. 1. Adhesion-dependent integrin signaling, not low WSS magnitude per se, is the necessary hemodynamic trigger for NLRP3 priming. 2. Assembly is rate-limited by dwell time, so in-tissue cells exceed transmigrating cells despite lower NF-κB, an ordering that no single model rule produces. 3. Recruitment and priming are spatially separated: adhesion peaks in the aneurysmal segment, whereas per-cell priming is 1.81-fold higher at the bifurcation than at the inlet, although that gradient is prescribed. The biological model is not calibrated against experiment; validation of the first two is planned by staging adhesion duration on an endothelial monolayer and counting ASC specks with a fluorescent reporter [5]. The framework is now being extended to a closed-loop circulation linking the lumbar discs and an osteoarthritic knee to test whether mediators released at one site affect the other.

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